Memory Circuits & Systems

Exercise 5

Ramulator2.0: DRAM Controller Simulator

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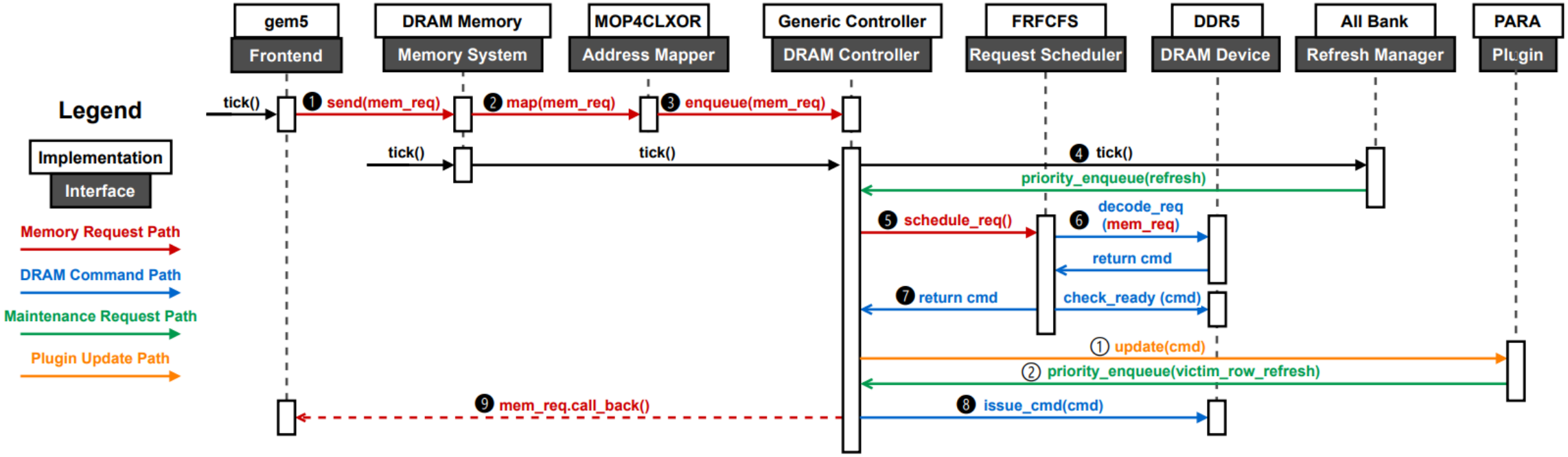
What is Ramulator2.0

- Cycle-level DRAM controller simulator
- Ramulator2.0 Can Model
  - Memory requests from traces
  - DRAM hierarchy
  - Memory controller behavior
  - DRAM timing constraints
  - Row buffer behavior
  - Performance statistics

Ramulator2.0 Simulation Flow



Setup

- tar -xvf /RAID2/COURSE/2026_Spring/es26mcs/es26mcsTA05/MCS_EX5.tar
- cd ramulator2
- source .cshrc (Ramulator2.0 needs cmake & new version of g++)

```
20:43 es26mcsTA04@ee31[~/MCS_EX5_YMM/final/ramulator2]$ source .cshrc

In MCS Ramulator HW environment, do not type the keywords yes', 'layout' or 'coding'
```

- Run ./00_setup.tcl

```
1   mkdir ./build
2   mkdir ./traces
3   mkdir ./traces_log
4   mkdir ./cmd_records
5   mkdir ./returned_trace
6   mkdir ./sent_request_trace
7   cp run_ramulator ./build/01_run_ramulator
```

- Note: the ./09_clean_up.tcl will clean all the log file and compile file, please use it carefully

Prepare file

- cd ./scripts to gen the trace file
  - python3 generate_trace.py --pattern both --st-count 256 --ld-count 256 --addr-space 1048576 --out-prefix trace
- Trace file

trace_sequential.trace

```
ST 0x0 0 0
ST 0x80 10 0
ST 0x100 0 0
ST 0x180 0 0
ST 0x200 0 0
ST 0x280 0 0
ST 0x300 0 0
ST 0x380 0 0
ST 0x400 0 0
ST 0x480 0 0
ST 0x500 0 0
ST 0x580 0 0
ST 0x600 0 0
ST 0x680 0 0
ST 0x700 0 0
ST 0x780 0 0
```

- Config file

config.yaml

```
Frontend:
  impl: LoadStoreStallTrace
  clock_ratio: 3
  num_expected_insts: 512
  returned_trace_path: ../returned_trace/
  trace_served_file_path: ../sent_request_trace/
  bandwidth_sample_time_interval: 100
  debug: false
  traces:
    - ../traces/trace_sequential.trace

  Translation:
    impl: NoTranslation
    max_addr: 33554432
```

You need to change the trace file path right here

Compile

- cd build
- cmake ..

```
20:43 es26mcsTA04@ee31[~/MCS_EX5_YMM/final/ramulator2/build]% cmake ..
-- The CXX compiler identification is GNU 13.2.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /RAID2/COURSE/2026_Spring/es26mcs/es26mcsTA05/share/local_tools/gcc-13.2.0/bin/g++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
Configuring Ramulator ...
Configuring yaml-cpp...
CMake Deprecation Warning at ext/yaml-cpp/CMakeLists.txt:2 (cmake_minimum_required):
  Compatibility with CMake < 3.5 will be removed from a future version of
  CMake.

  Update the VERSION argument <min> value or use a ...<max> suffix to tell
  CMake that the project does not need compatibility with older versions.


Done configuring yaml-cpp.
Configuring spdlog...
-- Build spdlog: 1.11.0
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Failed
-- Looking for pthread_create in pthreads
-- Looking for pthread_create in pthreads - not found
-- Looking for pthread_create in pthread
-- Looking for pthread_create in pthread - found
-- Found Threads: TRUE
-- Build type: Release
Done configuring spdlog.
Configuring argparse...
Done configuring argparse.
-- Configuring done (20.8s)
-- Generating done (2.3s)
-- Build files have been written to: /RAID2/COURSE/2026_Spring/es26mcs/es26mcsTA04/MCS_EX5_YMM/final/ramulator2/build
```

- make –j8

```
20:43 es26mcsTA04@ee31[~/MCS_EX5_YMM/final/ramulator2/build]% make -j8
[  8%] Building CXX object _deps/spdlog-build/CMakeFiles/spdlog.dir/src/spdlog.cpp.o
[  8%] Building CXX object src/translation/CMakeFiles/ramulator-translation.dir/impl/no_translation.cpp.o
[  8%] Building CXX object src/base/CMakeFiles/ramulator-base.dir/factory.cpp.o
[  8%] Building CXX object src/test/CMakeFiles/ramulator-test.dir/test_impl.cpp.o
[  8%] Building CXX object src/memory_system/CMakeFiles/ramulator-memorysystem.dir/impl/Global_Controller.cpp.o
```

```
[ 96%] Building CXX object src/dram_controller/CMakeFiles/ramulator-controller.dir/impl/plugin/prac/prac.cpp.o
[ 96%] Built target ramulator-controller
[ 97%] Linking CXX shared library /RAID2/COURSE/2026_Spring/es26mcs/es26mcsTA04/MCS_EX5_YMM/final/ramulator2/libramulator.so
[ 97%] Built target ramulator
[ 98%] Building CXX object CMakeFiles/ramulator-exe.dir/src/main.cpp.o
[100%] Linking CXX executable ramulator2
[100%] Built target ramulator-exe
```

Run Ramulator2.0

- ./01_run_ramulator ../config_files/config.yaml

Open

01_run_ramulator
~/MCS_EX5_YMM/final/ramulator2/build

```
#!/bin/bash

CONFIG=$1
LOG=${2:-../traces_log/run.log}

./ramulator2 -f "$CONFIG" > "$LOG" 2>&1
```

```
22:04 es26mcsTA04@ee31[~/MCS_EX5_YMM/final/ramulator2/build]$ ./01_run_ramulator ../config_files/config.yaml
```

- The result will store in ../traces_log/run.log

Open Save

run.log
~/MCS_EX5_YMM/final/ramulator2/traces_log

```
Frontend:
  impl: LoadStoreStallTrace
  average_bandwidth: 47.6279068
  worst_bandwidth: .inf
  peak_bandwidth: -1

MemorySystem:
  impl: Global_Controller
  bandwidth_utilization: 0  # The bandwidth utilization in percentage
  worst_bandwidth: 0
  peak_bandwidth: 0
  average_bandwidth: 0
  total_num_other_requests: 0
  total_num_write_requests: 256
  total_num_read_requests: 256
  memory_system_cycles: 688
```

TraceRecorder Plugin

- Check DRAM timing behavior
- Observe row hit / row conflict
- Understand row policy behavior
- Explain scheduling decisions

```yaml
- ControllerPlugin:
    impl: TraceRecorder
    path: ../cmd_records/cmd_records.trace
```

In src/dram/impl/DDR4.cpp

```cpp
inline static constexpr ImplDef m_levels = {
  "channel", "rank", "bankgroup", "bank", "row", "column",
};
```

cmd_record.ch0

```
3, ACT, 0, 0, 0, 0, 0, 0
10, WR, 0, 0, 0, 0, 0, 0
14, WR, 0, 0, 0, 0, 0, 1
18, WR, 0, 0, 0, 0, 0, 2
22, WR, 0, 0, 0, 0, 0, 3
26, WR, 0, 0, 0, 0, 0, 4
30, WR, 0, 0, 0, 0, 0, 5
34, WR, 0, 0, 0, 0, 0, 6
38, WR, 0, 0, 0, 0, 0, 7
42, WR, 0, 0, 0, 0, 0, 8
46, WR, 0, 0, 0, 0, 0, 9
50, WR, 0, 0, 0, 0, 0, 10
54, WR, 0, 0, 0, 0, 0, 11
58, WR, 0, 0, 0, 0, 0, 12
62, WR, 0, 0, 0, 0, 0, 13
66, WR, 0, 0, 0, 0, 0, 14
70, WR, 0, 0, 0, 0, 0, 15
86, PRE, 0, 0, 0, 0, 1, 0
```

1.Address mapping (50%)

- DRAM Architecture in this lab
  - 4 channels, 1 bank per channel, 65536 rows per bank, 16 columns per row
- Timing Constraint determine when a DRAM command can be issued
  - tRCD: ACT → READ/WRITE delay
  - tRP: PRE → Next ACT delay
  - tCL: Time from READ command to data available
  - tRAS: Minimum row active time
  - tRC: Minimum time between two ACTs to the same bank
  - tRFC: Time required to complete a refresh operation
- DRAM configuration
  - File: src/dram/impl/DDR4.cpp

```
    //    name           rate           nBL                nCL                  nRCD                 nRP             nRAS               nRC          nWR          nRTP          nCWL(TSV as
IO)  nCCDS nCCDL nRRDS nRRDL nWTRS nWTRL nFAW  nRFC nREFI nCS,  tCK_ps
     {"DDR4_3DDRAM_1024", {2000,        2,                   5,                   7,                   3,              10,                12,          9,           8,
5,           4,    4,   -1,    -1,   8,     8,  -1,   -1,   -1,  2,   1000}},
```

1.Address mapping (50%)

- Q1-1: Analysis with Given Address Mapping
  - Run **sequential** and **random** traces using the provided address mapping.
  - Analyze and explain
    - The issued DRAM command sequence (ACT / READ / PRE)
    - Why each command is issued
    - Which timing constraints delay the next command

1.Address mapping (50%)

- Q1-2: Design a new address mapping scheme to improve memory performance.
  - Modify **src/addr_mapper/impl/linear_mappers.cpp** to change the address mapping scheme.
    - Hint: modify the different order for Ch/Ra/Ba/Row/Col
    - **You need to compile again!!**
  - Analysis how to increase the effective memory access bandwidth.

```
class New_mapping final : public LinearMapperBase, public Implementation {
  RAMULATOR_REGISTER_IMPLEMENTATION(IAddrMapper, New_mapping, "New_mapping", "Applies a New_mapping mapping to the address.");

  public:
    void init() override { };

    void setup(IFrontEnd* frontend, IMemorySystem* memory_system) override {
      LinearMapperBase::setup(frontend, memory_system);
    }

    void apply(Request& req) override {
      req.addr_vec.resize(m_num_levels, -1);
      Addr_t addr = req.addr >> m_tx_offset;

      // TO DO HERE

      }
    }
};
```

```
inline static constexpr ImplDef m_levels = {
  "channel", "rank", "bankgroup", "bank", "row", "column",
};
```

2.Row-policy (50%)

- Row-Buffer Behavior
  - A DRAM bank can keep only one row active at a time
  - Accessing the currently open row results in a Row Hit
  - Accessing a different row causes a Row Conflict, requiring PRECHARGE + ACTIVATE operations.
- Open-row policy
  - Keeps the active row open after a memory access
  - Increases the probability of future **row hits**
- Close-row policy
  - Closes the active row after a certain number of accesses
  - Reduces row conflicts for workloads with low row locality or random accesses
- File Path: src/dram_controller/impl/rowpolicy/basic_rowpolicies.cpp
- Cap: Maximum number of column accesses allowed to the same row in Close row policy

```
RowPolicy:
  impl: OpenRowPolicy
  cap: 4
```

```
RowPolicy:
  impl: ClosedRowPolicy
  cap: 4
```

2.Row-policy (50%)

- Q2-1: Row-Policy Analysis
  - Run the sequential and random traces and compare the behavior of open-row and close-row policies.
    - Any address mapping scheme may be used. Please clearly state the mapping used in your experiments.
  - Analyze and report
    - Memory bandwidth, Number of row hits, Number of row conflicts
    - DRAM command distribution (ACT / WRITE / READ / PRE)
  - Which traces type is suitable for open-row which for close-row

Report notice

- In your analysis please provide evidence to support your reason like log_file screenshot, code modify screenshot, etc.

Submission on E3 platform

- Please compress your **report** & **linear_mappers.cpp or another code you added in a single compressed file(.zip)** and upload this single file on E3 platform
  - **Report: Simulation result, comparison, analysis and the evidence picture.**
  - Naming rule: Ex5_#ID.zip (including report & code)
- Due date: 6/23 P.M 23:59